Grothendieck Ring Classification
Current Research Status and Audit Corrections

Fusion Frontier research project

24 October 2026

AI-generated research; independent human verification and formalization pending. All theorem and classification labels describe the project's current claims. Exact finite replays do not prove their external categorical premises, and a verified tensor list alone does not prove enumeration completeness.

1 Current classification scope

Ranks two, three and four have unrestricted classification claims, with two, five and fifteen realized based rings respectively. Higher ranks have the explicit bounds below. Realization means existence of at least one model, not classification of all fusion categories up to tensor equivalence.

Rank	Through multiplicity	Excluded	Realized	Open
5	20	40982	17	0
6	9	25362	57	2
7	5	9976	30	0
8	3	4403	93	4
9	2	2145	76	4
10	1	136	58	0
11	1	155	27	0

There are **10 current generally unresolved cases**. The exact current list and tensors are supplied in `data/current_remaining_cases.py`. Rank six is complete through multiplicity three; rank eight through two; rank nine through one. These finite statements are relative to the supplied census and the recorded categorification decisions.

2 Current theoretical results

The ten open cases are listed in rank order:

- Rank six: `R6_4_155`, `R6_8_1660`.
- Rank eight: `R8_3_928`, `R8_3_941`, `R8_3_1154`, `R8_3_1768`.
- Rank nine: `R9_2_137`, `R9_2_1607`, `R9_2_1975`, `R9_2_1976`.

No new exclusion of one of these source rings is claimed.

Rank ten is complete at multiplicity one. The classical conformal embedding $SU(4)_6 \subset SU(10)_1$ supplies an actual connected etale algebra $B = 000 + 060 + 202 + 222$. The charge-zero component of its module category realizes $R10_1_189$. The proof uses minimal ground-state branching and dimension exhaustion; all 1000 census coefficients agree. An actual rank-fourteen modular category and its algebra are also constructed. This recognizes a known construction; no new tensor category is claimed.

Rank eight retains four open cases. R928 has categorical Morita rigidity without a module-rank cutoff. For R1768 an actual complex twisted $GH(C_6, 2)$ precursor has now been constructed with precisely the required pointed associator. The connected separable algebra $1 + \sigma$ needed for the source ring remains unproved. The canonical inherited algebra is instead $\mathrm{End}(\sigma)$. Its remaining multiplication equations are explicit in two scalar variables. Five pointed-extension construction families for R1154 and further exceptional quantum-group routes are ruled out; those route obstructions do not imply general nonexistence.

Rank nine retains four open cases. The centers have no nontrivial Tannakian subcategory, and their integral simples are invertible. A new module-functor sign theorem gives local Clifford restrictions on each large object in the three commutative sources. An all-cochain D8 theorem gives an aggregate condition for all four. Exact discrete certificates force projected cubic indicators $1/2$ in the positive-TY R137 and negative-TY R1976 branches, hence 9 divides their FS exponents. Exact feasible witnesses to the necessary systems remain; they are not categories.

Rank six retains two open cases. The even cyclic Clifford theorem passes all 1144 tested exact blocks for R155. The R1660–R941 identification is exact, but its categorical Morita link still requires the indicated nonpointed algebra.

The current proofs, independent agent audits and portable replays are in `evidence/unresolved_retry_20261024/`. The thirteen new portable checks pass, also with assertions disabled. They do not replace human verification of the categorical arguments.

3 R436: an independent replacement proof

The old TPE scalar-table completeness premise remains withdrawn. Its complete-table derivation assumes unitarity and does not parameterize an arbitrary original spherical category. Exact scalar computations cannot repair this categorical gap.

A separate spherical-equivariantization and formal-codegree argument now excludes $R8_3_436$ generally, without the unsupported parameterization. It is proved in `evidence/r436_repair_20261024/`. The conservative audit correction and the independent replacement are preserved as distinct research steps; the former is not presented as a valid general proof. Rank eight at exact multiplicity three therefore has 3507 exclusions, 16 unitary models and four open cases, and cumulatively 4403 exclusions, 93 models and four open cases.

4 Other corrections

The rank-five current replay is

```
python evidence/prepublication_20261017/rank5/replay.py --jobs 2
```

It checks all 40999 inputs, 40982 general exclusions and 17 models. The historical standalone

verifier leaves the two inert-prime frontiers unresolved in general and has complete bound seventeen. The discovery example is a bounded attempt that may return unresolved; it is not guaranteed fresh discovery of the stored composite certificate.

The signed rank-five commutativity lemma now explicitly requires the dimension character to be nonzero on every basis element. The pair-product Gram matrix's row count explicitly denotes the remaining central simple objects. The rank-seven bibliography now names P. G. Budinski. The census documentation updates the supported rank range, companion-PDF count and supplied, hash-matched rank-seven producer source; independent full re-enumeration remains separate. Older six-case frontier totals are scoped to their historical lower-rank range.

5 Completed audit checks and remaining limits

The audit checked all 1431327 tensors of its pinned census snapshot with the supplied exact verifier and used new independent small-bound enumeration and high-rank isomorphism programs. Its full rank-four, rank-five and rank-eleven classification replays passed. It checked 24383 of 25362 rank-six exclusions; 979 remain outside completed replay coverage. Selected rank-seven/eight proofs and complete hard certificates passed their stated checks. The audit's tensor total predates the separately supplied rank-nine multiplicity-two extension.

Large highest-bound enumerations were not all independently repeated. Every external categorical theorem and model associator was not independently reproved. The Evans–Gannon modular-data identification used by the R820/R869 models retains an outstanding error-bound verification concern; the audit did not establish a wrong model. No claim is made that every possible error has been found.

6 Reading and reproducing the atlas

The common record directory, unresolved directory and model catalogue offer exact structural filters for integral, weakly integral, simple, perfect, pointed, commutative, self-dual and multiplicity-free rings. The filters concern the underlying ring; they are independent of categorical existence. Rational root isolation decides integrality, while complete generated-subring closure decides simplicity. Aggregate rank–multiplicity counts retain their full recorded populations. Definitions, examples and standard references are on the atlas's theory page.

```
python tools/audit_corrections_20261024.py
python tools/verify_audit_revision.py
```

These commands reconcile the audit, the independent repair and the accepted unitary model, and validate current publication identities and structural data. They do not claim a new full replay of every historical categorical proof.

https://sebastienpalcoux.github.io/grothendieck-ring-classification/theory.html